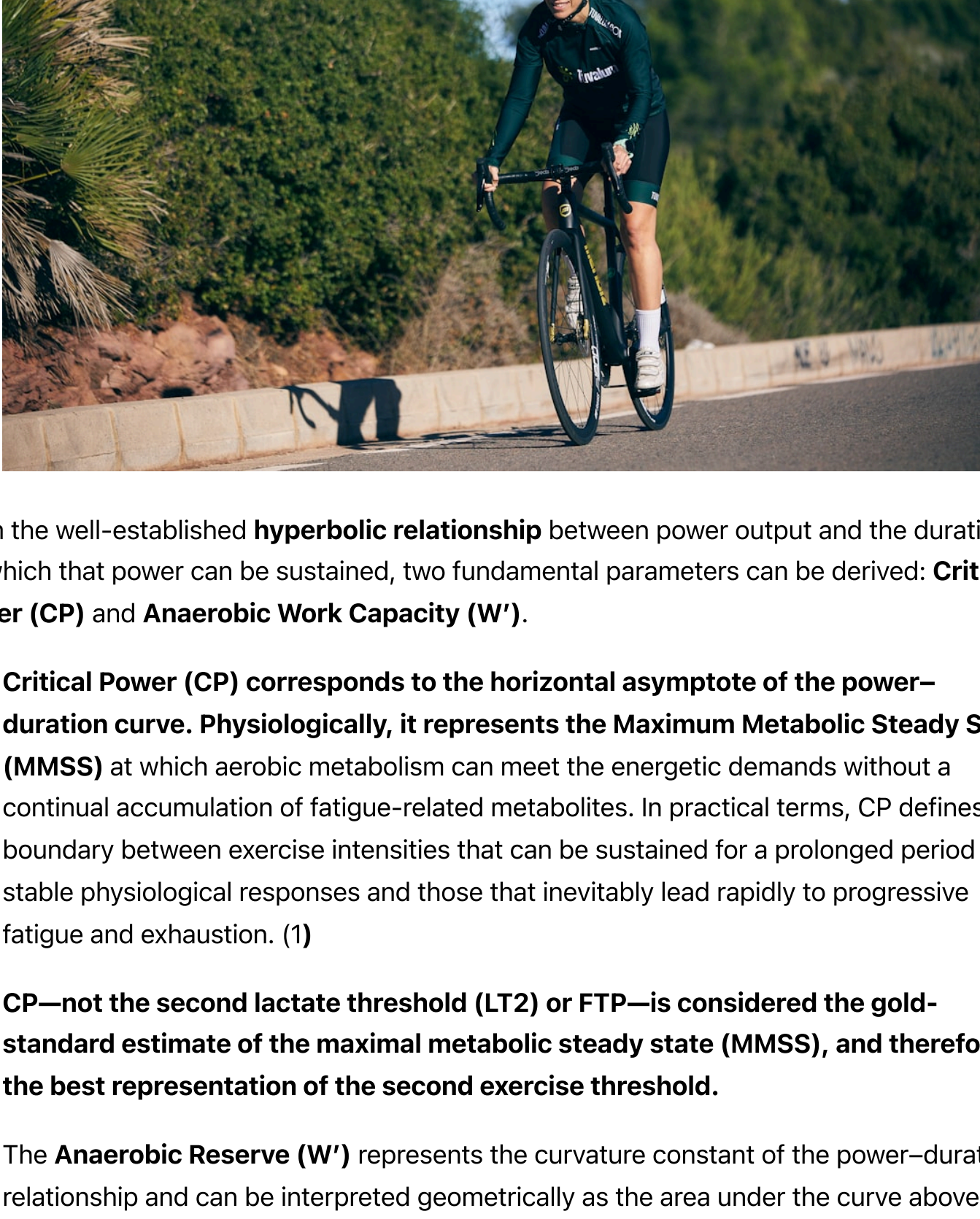


140. HOW TO ACCURATELY ESTIMATE CP AND W': THE 'BEST INDIVIDUAL FIT' METHOD



From the well-established **hyperbolic relationship** between power output and the duration for which that power can be sustained, two fundamental parameters can be derived: **Critical Power (CP)** and **Anaerobic Work Capacity (W')**.

- **Critical Power (CP)** corresponds to the horizontal asymptote of the **power–duration curve**. Physiologically, it represents the **Maximum Metabolic Steady State (MMSS)** at which aerobic metabolism can meet the energetic demands without a continual accumulation of fatigue-related metabolites. In practical terms, CP defines the boundary between exercise intensities that can be sustained for a prolonged period with stable physiological responses and those that inevitably lead rapidly to progressive fatigue and exhaustion. (1)

CP—not the second lactate threshold (LT2) or FTP—is considered the gold-standard estimate of the maximal metabolic steady state (MMSS), and therefore the best representation of the second exercise threshold.

- The **Anaerobic Reserve (W')** represents the curvature constant of the power–duration relationship and can be interpreted geometrically as the area under the curve above Critical Power at any given point. Physiologically, it corresponds to a finite amount of work (expressed in joules) that can be performed above CP.

Importantly, the magnitude of W' is theoretically fixed and independent of the specific power output selected above CP. While exercising at a higher intensity accelerates the rate at which W' is depleted, the total amount of work that can be completed above CP remains the same. **W' is considered a parameter reflecting anaerobic qualities and the capacity to sustain high-intensity efforts above CP.** (1)

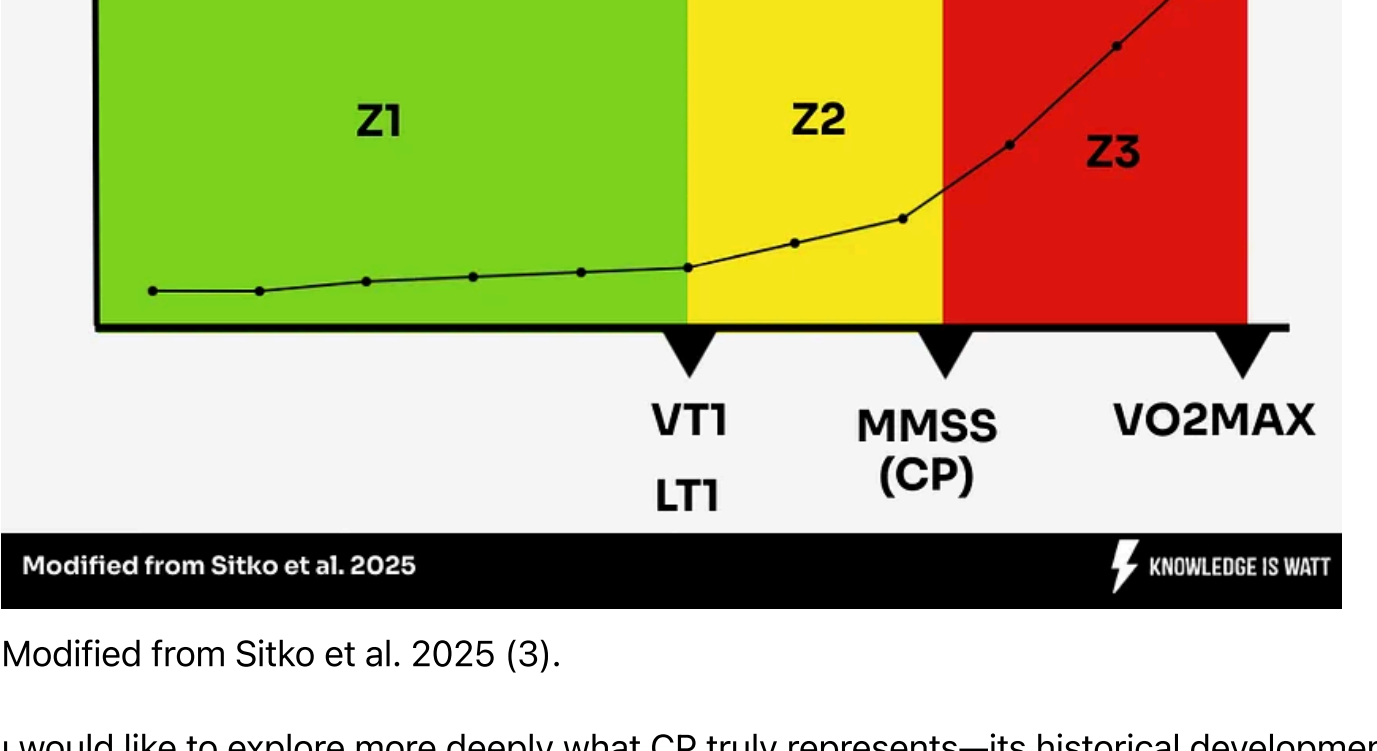
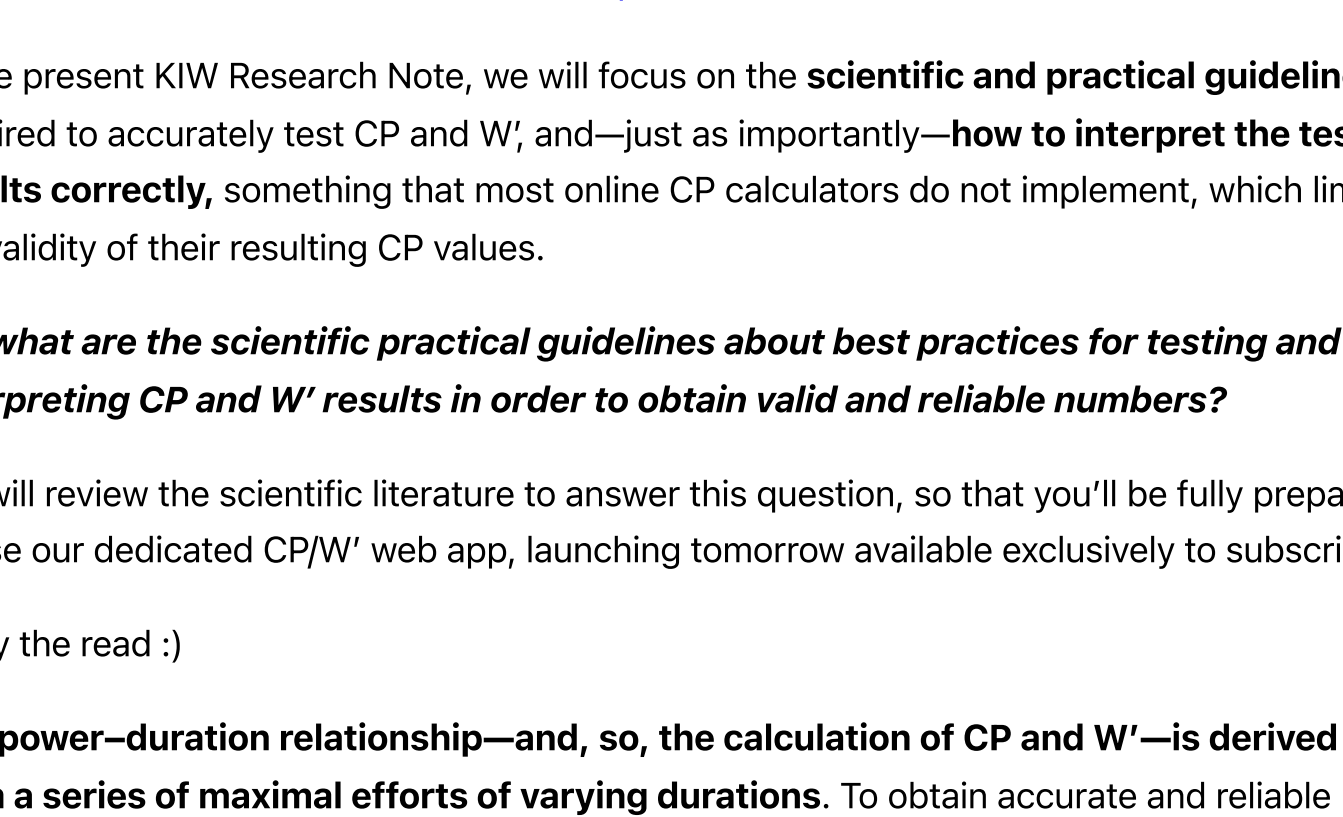


Image modified from Poole et al. 2016 (2)



Modified from Sitko et al. 2025 (3).

If you would like to explore more deeply what CP truly represents—its historical development, its physiological meaning, and why it should be considered the gold standard for identifying the maximum metabolic steady state (MMSS) and defining training zones—you can refer to [our dedicated KIW Research Note on the topic](#).

In the present KIW Research Note, we will focus on the **scientific and practical guidelines** required to accurately test CP and W', and—just as importantly—**how to interpret the test results correctly**, something that most online CP calculators do not implement, which limits the validity of their resulting CP values.

So, what are the scientific practical guidelines about best practices for testing and interpreting CP and W' results in order to obtain valid and reliable numbers?

We will review the scientific literature to answer this question, so that you'll be fully prepared to use our dedicated CP/W' web app, launching tomorrow available exclusively to subscribers.

Enjoy the read :)

The power–duration relationship—and, so, the calculation of CP and W'—is derived from a series of maximal efforts of varying durations. To obtain accurate and reliable estimates of CP and W', it is essential to follow specific guidelines when performing these maximal efforts. Key factors to consider are:

- **Number of efforts**
- **Duration and Intensity of each effort**
- **Recovery between efforts**
- **Terrain**
- **Cycling Position**
- **Cadence**
- **Pacing Strategy**

Typically, **three maximal efforts** are considered sufficient to construct the hyperbolic power–duration relationship and calculate CP and W' with good accuracy. (1) Please note that with only two efforts it is not possible to assess the accuracy of the CP/W' estimates as metrics such as the standard error of the estimate (SEE) and the *best individual fit* method (described in the next sections) cannot be applied. Therefore, **we recommend performing three trials.**

All efforts should be performed in the **severe intensity domain (above CP)**. It is **recommended to set the shortest trial at 2–3 minutes and the longest trial at 12–15 minutes**. This range ensures that the power–duration curve is well covered, allowing a good fit while avoiding skewness. **The shortest effort should not be shorter than 2–3 minutes**; if it is too brief, the intensity may be so high that fatigue occurs before reaching **VO₂max** (maximal oxygen uptake) and it would represent a type of effort not covered by the CP-W' power duration-relationship (extreme intensity domain). Conversely, **the longest effort should not exceed 12–15 minutes**, as longer durations increase the influence of glycogen depletion, motivation, and pacing strategies on the results. (1, 4) Given these considerations, a good scientific practical approach can be:

- 1st effort: 3 minutes
- 2nd effort: 8 minutes
- 3rd effort: 12-13 minutes

"Inter-trial recovery between prediction trials should be set to a minimum of 30 min during a single visit or 24 h during multiple days. The benefit of multiple days if that any fatigue induced by the initial prediction trial does not affect the subsequent one, but possible error due to day-to-day variation in power output is introduced".

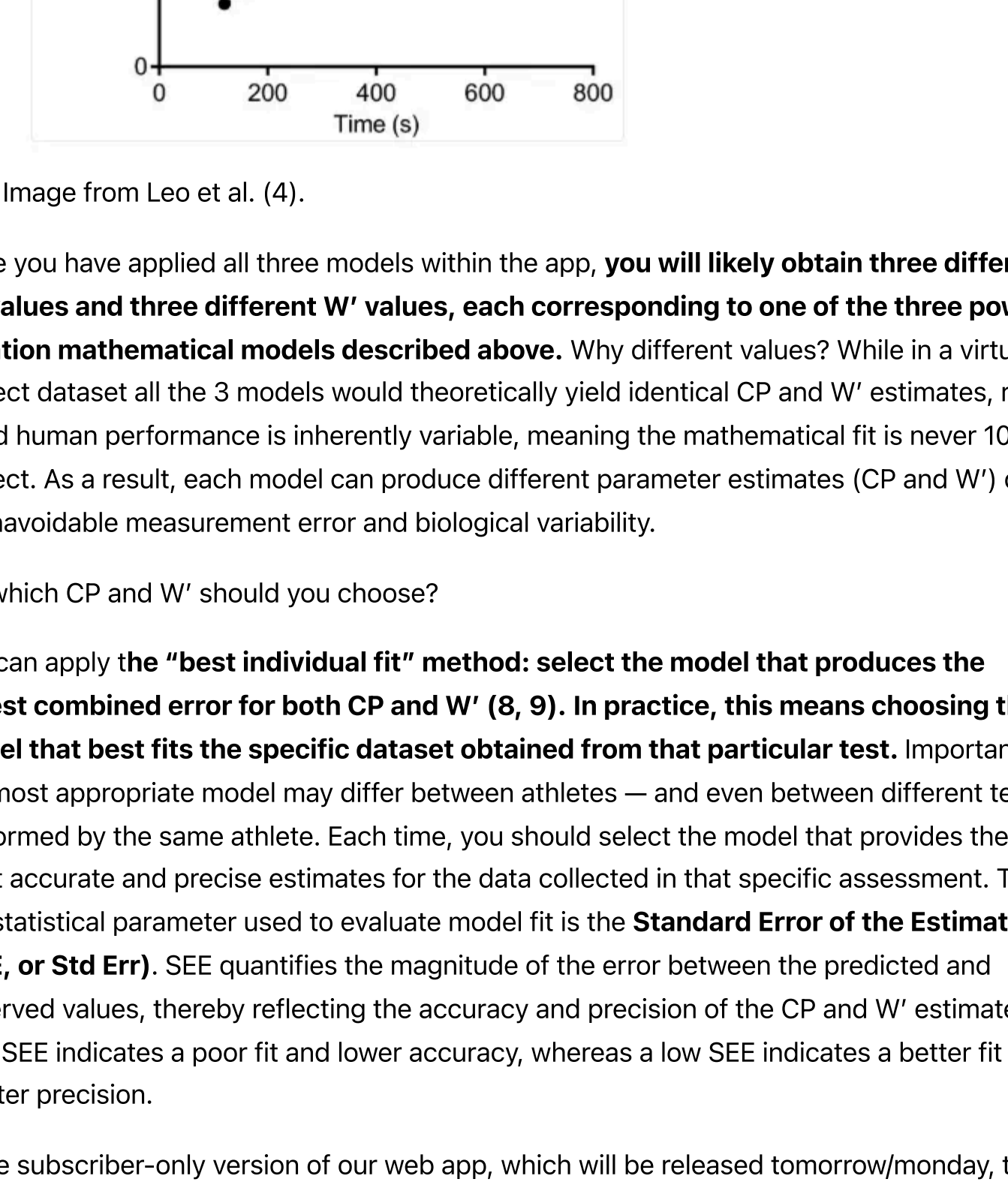
– Leo et al. 2022 – (5)

Another factor to be considered is whether the all-out efforts should be performed on **flat or uphill terrain**. A study reported that mean maximal power outputs across different durations are on average about **3% higher uphill** (6–7% gradient) compared with flat terrain. (5) So, the advice is to perform **all trials on similar terrain**. Which one? **The type where you train and where you want to be most competitive**: for example, uphill for climbers and flat for classics specialists. When applying the resulting training zones to a different terrain, just adjust them by approximately **3-5% above or below** to account for the difference relative to the test conditions.

Cycling position (standing, seated, or time trial (TT) position) can significantly affect mean maximal power output during all-out efforts. So, **CP and W' values obtained in different positions should not be used interchangeably**. For example, likely due to the more compressed position on the bike, CP measured in a time trial position is usually lower than on a standard road bike, (6) with the magnitude of the difference depending likely on the aggressiveness of the TT position and the rider's experience on a TT bike. **When performing a CP test on a road bike, it is recommended to use your preferred position** (seated, standing, or alternating), as this will most closely reflect how you ride in training and races, making the results more valid and practically useful.

Cadence should be kept consistent across all trials, as CP calculations are influenced by pedalling rate. (7) A useful guideline is to **aim for the same cadence within ±5 rpm** for all efforts. In practice, this consistency in cadence across different trial is often achieved naturally without giving any specific indications to the riders.

You should aim for an optimal even pacing rather than starting all-out and fading. A good practical approach is to begin with a fast but controlled start of about 30 seconds—around +10% faster but not all-out—to help accelerate the rise of oxygen consumption. Then maintain an even pacing (at your best) throughout the effort, trying to increase power in the final 30 seconds (if possible). Of course, to attain an optimal even pacing, it is crucial to **choose a route free from interruptions—no traffic lights, traffic jams, strong wind, and a constant gradient**.



After completing the trials, you should have collected your results: power output and duration data. Below is an example of how these results might look:

	Power (Watt)	Duration (sec)
Effort 1	540	180
Effort 2	430	360
Effort 3	400	720

Using an Excel spreadsheet or a web-based calculator (tomorrow/on monday paid subscribers will get access to our new scientific practical CP/W' web app version!), you should then plot the power and duration values obtained from these trials **according to each of the following three mathematical models**:

- **The 2-Parameter Hyperbolic Model (CP 2-P Hyp)**: In this model, CP represents the asymptote of the power–duration curve, while W' is the curvature constant, corresponding to the finite amount of work that can be performed above CP (i.e., the area under the curve above CP at any given point). See Panel a below.
- **The 2-Parameter Linear Power–Inverse Time Model (CP 2–1/time)**: In this case, W' is represented by the slope of the linear relationship, whereas the y-intercept corresponds to CP. See Panel b below.
- **The 2-Parameter Linear Work–Time Model (CP 2-linear)**: In this model, W' is represented by the slope of the line, and CP corresponds to the y-intercept. See Panel c below.

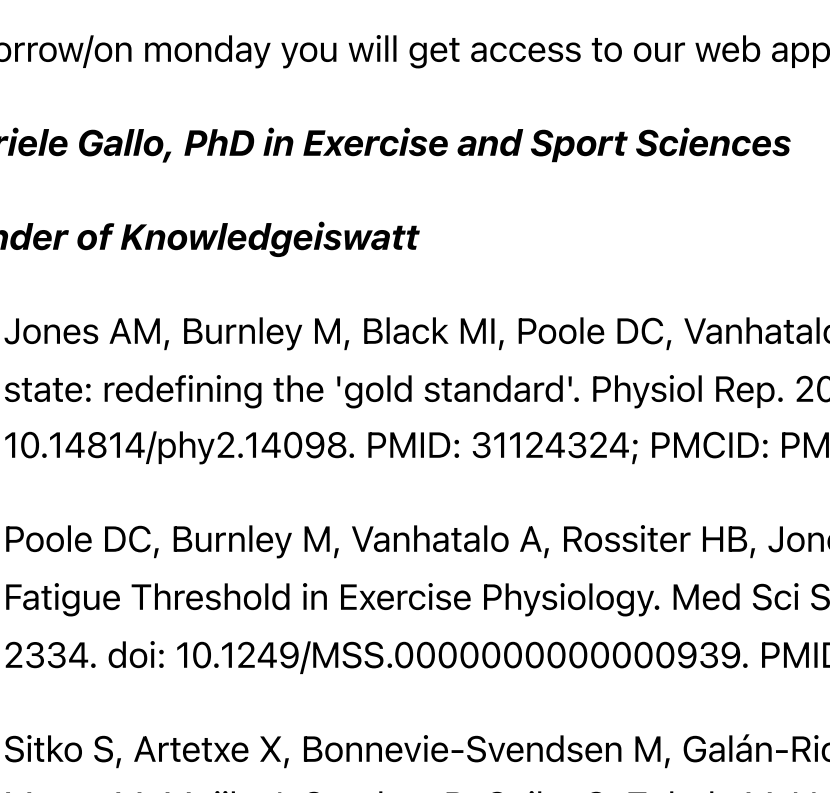


Image from Leo et al. (4).

Once you have applied all three models within the app, **you will likely obtain three different CP values and three different W' values, each corresponding to one of the three power–duration mathematical models described above**. Why different values? While in a virtual perfect dataset all the 3 models would theoretically yield identical CP and W' estimates, real-world human performance is inherently variable, meaning the mathematical fit is never 100% correct. As a result, each model can produce different parameter estimates (CP and W') due to unavoidable measurement error and biological variability.

So, which CP and W' should you choose?

You can apply the **“best individual fit” method: select the model that produces the lowest combined error for both CP and W' (8, 9)**. In practice, this means **choosing the model that best fits the specific dataset obtained from that particular test**. Importantly, the most appropriate model may differ between athletes — and even between different tests performed by the same athlete. Each time, you should select the model that provides the most accurate and precise estimates for the data collected in that specific assessment. The key statistical parameter used to evaluate model fit is the **Standard Error of the Estimate (SEE, or Std Err)**. SEE quantifies the magnitude of the error between the predicted and observed values, thereby reflecting the accuracy and precision of the CP and W' estimates. A high SEE indicates a poor fit and lower accuracy, whereas a low SEE indicates a better fit and greater precision.

In the subscriber-only version of our web app, which will be released tomorrow/monday, the recommended model (and so CP and W') is automatically highlighted in green. **This corresponds to the model with the lowest combined parameter uncertainty**—that is, the combination of CP and W' standard error of estimate (SEE).

Most online CP calculators do not implement this procedure, which limits the validity of their resulting CP values.

Below is an example from our web app using 'real-world' data of a test performed last year. In this case, selecting the wrong model could have resulted in underestimating CP by approximately 20 W (313 W vs. 330 W) This clearly illustrates the importance of adopting the “best individual fit” approach.

Best Individual Fit Method					
Model	CP (W)	W' (J)	SEE CP	SEE W'	Combined Parameter Uncertainty
Linear Power-Inverse Time Model	330.4	19.05	10.19 (3.1%)	2.90 (15.2%)	10.59
Linear Work-Time Model	323.8	21.72	12.27 (3.8%)	6.26 (28.8%)	13.77
2-Parameter, Hyperbolic Model	312.6	28.08	12.99 (4.2%)	8.39 (29.9%)	15.47

When should you consider repeating one or more efforts because no model fits your data well enough?

- You should consider performing another trial to **improve the accuracy and precision of CP if, in all three models, the Standard Error of the Estimate (SEE or Std Err) for CP exceeds 5% (1)**.
- Similarly, you should consider performing another trial to **improve the accuracy and precision of W' if, in all three models, the SEE for W' exceeds 10% (1)**.

Keep in mind that these thresholds are not absolute. For example, if the CP SEE is **6%**, it may still be acceptable, but if it reaches **7.5–10%**, you can consider performing another effort. A higher SEE indicates that, in one or more trials, pacing may have been suboptimal or the athlete just underperformed. In such cases, it is suggested to repeat the exact trial(s) that were not performed optimally.

Thanks for Reading and for Being Part of this Journey!

Share with your Team!

Tomorrow/on monday you will get access to our web app!

Gabriele Gallo, PhD in Exercise and Sport Sciences

Founder of Knowledgeiswatt

1. Jones AM, Burnley M, Black MI, Poole DC, Vanhatalo A. The maximal metabolic steady state: redefining the 'gold standard'. *Physiol Rep*. 2019 May;7(10):e14098. doi: 10.14814/phy2.14098. PMID: 31124324; PMCID: PMC6533178.
2. Poole DC, Burnley M, Vanhatalo A, Rossiter HB, Jones AM. Critical Power: An Important Fatigue Threshold in Exercise Physiology. *Med Sci Sports Exerc*. 2016 Nov;48(11):2320-2334. doi: 10.1249/MSS.0000000000000939. PMID: 27031742; PMCID: PMC5070974.
3. Sitko S, Artetxe X, Bonnevie-Svendsen M, Galán-Rioja MÁ, Gallo G, Grappe F, Leo P, Mateo M, Mujika I, Sanders D, Seiler S, Zabala M, Valenzuela PL, Viribay A. What Is "Zone 2 Training"? Experts' Viewpoint on Definition, Training Methods, and Expected Adaptations. *Int J Sports Physiol Perform*. 2025 Feb 26;20(11):1614-1617. doi: 10.1123/ijsp.2024-0303. PMID: 40010355.
4. Leo P, Spragg J, Podlogar T, Lawley JS, Mujika I. Power profiling and the power-duration relationship in cycling: a narrative review. *Eur J Appl Physiol*. 2022 Feb;122(2):301-316. doi: 10.1007/s00421-021-04833-y. Epub 2021 Oct 27. PMID: 34708276; PMCID: PMC8783871.
5. Valenzuela PL, Mateo-March M, Muriel X, Zabala M, Lucia A, Pallares JG, Barranco-Gil D. Road gradient and cycling power: An observational study in male professional cyclists. *J Sci Med Sport*. 2022 Dec;25(12):1017-1022.
6. Kordi M, Fullerton C, Passfield L, Parker Simpson L. Influence of upright versus time trial cycling position on determination of critical power and W' in trained cyclists. *Eur J Sport Sci*. 2019 Mar;19(2):192-198. doi: 10.1080/17461391.2018.1495768. Epub 2018 Jul 14. PMID: 30009673.
7. Broxterman RM, Ade CJ, Barker T, Barstow TJ. Influence of pedal cadence on the respiratory compensation point and its relation to critical power. *Respir Physiol Neurobiol*. 2015 Mar;208:1-7. doi: 10.1016/j.resp.2014.12.008. Epub 2014 Dec 15. PMID: 25523595.
8. Black MI, Jones AM, Bailey SJ, Vanhatalo A. Self-pacing increases critical power and improves performance during severe-intensity exercise. *Appl Physiol Nutr Metab*. 2015 Jul;40(7):662-70. doi: 10.1139/apnm-2014-0442. Epub 2015 Mar 11. PMID: 26088158.
9. Black MI, Jones AM, Blackwell JR, Bailey SJ, Wylie LJ, McDonagh ST, Thompson C, Kelly J, Sumners P, Mileva KN, Bowtell JL, Vanhatalo A. Muscle metabolic and neuromuscular determinants of fatigue during cycling in different exercise intensity domains. *J Appl Physiol* (1985). 2017 Mar 1;122(3):446-459. doi: 10.1152/japplphysiol.00942.2016. Epub 2016 Dec 22. PMID: 28008101; PMCID: PMC5429469.